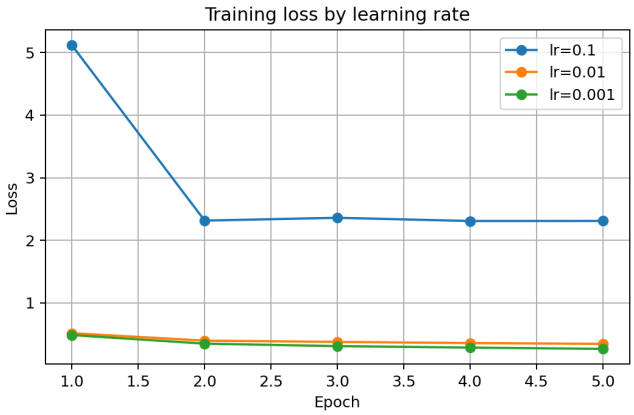
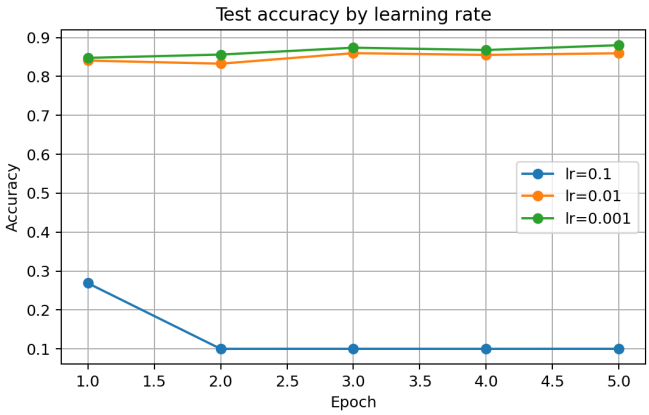
Week 2 Learning Rate Experiment

Fashion-MNIST MLP comparison

Experiment setup

I trained the same MLP on Fashion-MNIST with three learning rates: 0.1, 0.01, and 0.001. The model structure, batch size, optimizer, number of epochs, data, and random seed were kept the same. The network was 784 → 256 → 128 → 10 with ReLU activations and Adam optimization.





What I observed

The learning rate 0.1 was too large for this experiment. The loss became unstable and the test accuracy dropped to about 10%, which is close to random guessing for ten classes. The learning rate 0.01 learned quickly and reached about 86.0% test accuracy after five epochs. The learning rate 0.001 improved more steadily and reached about 88.0% test accuracy, although its first updates were slower.

For this short experiment, 0.001 gave the most stable and strongest final result. This conclusion only applies to the current model, optimizer, dataset, and five-epoch setting; a different setup could prefer another learning rate.

Short Answers

Learning rate questions

1. What happens when the learning rate is too large?

When the learning rate is too large, parameter updates can jump over good solutions. The loss may oscillate or fail to decrease, and the model may become unstable or perform close to random guessing.

2. What happens when it is too small?

When the learning rate is too small, each update is tiny. Training becomes slow, and the model may not reach a good result within the available number of epochs.

3. Why should we change only one variable when running this experiment?

Changing only the learning rate makes this a controlled experiment. If the model results change, I can reasonably connect the difference to the learning rate instead of being unsure whether the model, data, batch size, or another setting caused it.

Final note

This experiment helped me see that a training result depends not only on the model structure, but also on how the optimizer updates the parameters. The curves make the effect of the learning rate easier to understand than a single final accuracy number.